

MAMADOU BA

(347) 940 –2875 | Mamadou.Ba@CIX.CSI.CUNY.EDU | Staten Island, NY | [Linkedin](#) | [Github](#)

EDUCATION

City University of New York, College of Staten Island

Expected May 2027

Bachelor of Science in Computer Science

Relevant coursework: Data Structures (C++), Introduction to Data Science (Python), Computer Organization, Calculus II, Personal Financial Management

TECHNICAL SKILLS

- **Programming Languages:** Python (Pandas, NumPy), C/C++, SQL
- **Software/Tools:** Power Bi, MATLAB, Visual Studio, Jupyter Notebooks, Excel, Microsoft Copilot Studio
- **Spoken Languages:** English, French, Bambara (all fluent)

RELEVANT EXPERIENCE

Apex Forum | Hybrid Operations Analyst & AI Lead

March 2025 – Present

- Designed and deployed **AI automation workflows** with Microsoft Copilot Studio, eliminating **8+ hours of manual tasks weekly**
- Enhanced organizational visibility and engagement by producing analytics-driven insights on podcast and community initiatives, driving **30% growth in member participation**.
- Analyze operational processes to identify bottlenecks and implement AI-driven solutions, **improving team efficiency by 25%**
- Serve as **Microsoft 365 administrator**, managing Teams, SharePoint, and workflow integrations for distributed team collaboration

CodePath | Remote Software Engineering Fellow (Internship Connection Program)

June 2025 – August 2025

- Selected for an intensive software engineering fellowship focused on technical interview preparation and practical coding skills.
- Mastered algorithmic problem solving and data structures through rigorous weekly coding labs and technical challenges.

PROJECTS

Financial Market Data Analysis (Python)

November 2025 – Present

- Developed a Python script using **Pandas** and **yfinance** to extract historical stock data for S&P 500 companies.
- Implemented statistical analysis using **NumPy** to calculate moving averages, volatility, and daily returns to identify market trends.
- Visualized financial metrics using **Matplotlib** to generate comparative performance charts.

Algorithmic Trading Simulation (C++)

October 2025

- Built a trade execution engine in C++ utilizing hashmaps to efficiently store and retrieve stock ticker data.
- Simulated high-frequency order processing logic to understand latency and memory management in financial systems.

PROFESSIONAL & LEADERSHIP EXPERIENCE

Nike | Sales Associates | Staten Island, NY

February 2023 – March 2025

- Processed 100+ daily customer transactions using POS and inventory management systems, achieving 20% sales conversion rate
- Recognized as Employee of the Month (August 2023) for reliability and customer service in fast-paced retail environment

CSI Leadership Academy | Member | Staten Island, NY

September 2025 – Present

- Selected for competitive year-long leadership development program requiring 12 leadership workshops with 50 hours of volunteer service requirement

ASSOCIATIONS & INTERESTS

ColorStack @ CSI Computer Science Club | Member | Staten Island, NY

Interests: Podcast production and audio editing • Competitive soccer